DOT NET TECHNOLOGIES

UNIT -1

1.WHAT IS .NET?

- NET is a software framework which is designed and developed by Microsoft.
- The first version of the .Net framework was 1.0 which came in the year 2002.
- In easy words, it is a virtual machine for compiling and executing programs written in different languages like C#, VB.Net etc.
- It is used to develop Form-based applications, Web-based applications, and Web services.
- There is a variety of programming languages available on the .Net platform, VB.Net and C# being the most common ones.
- It is used to build applications for Windows, phone, web, etc.

What is a Web Service?

Client to Client - Clients can use XML Web Services to communicate data

Client to Server - Clients can send data to and receive data from servers.

Server to Server - Servers can share data with each other.

Service to Service - web services can work together.

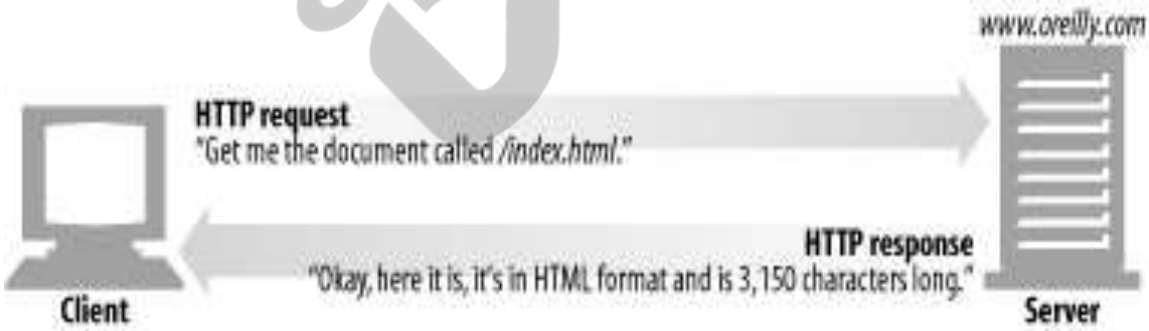
1.1 THE WEB CLIENT/ SERVER MODEL

The Web Clients and Web Servers:

- Web Based environment takes place between Two entities:

  1.A Web Client , which is the software that requests files, data and services

  2 A Web Server,which fulfills the client requests for contents



Protocols for Web Client/Server Communication:

The Web Client and Server send information using TCP/IP

- **TCP -** Transmission Control Protocol

  TCP is responsible for breaking data down into small packets before they can be sent over a network, and for assembling the packets again when they arrive.

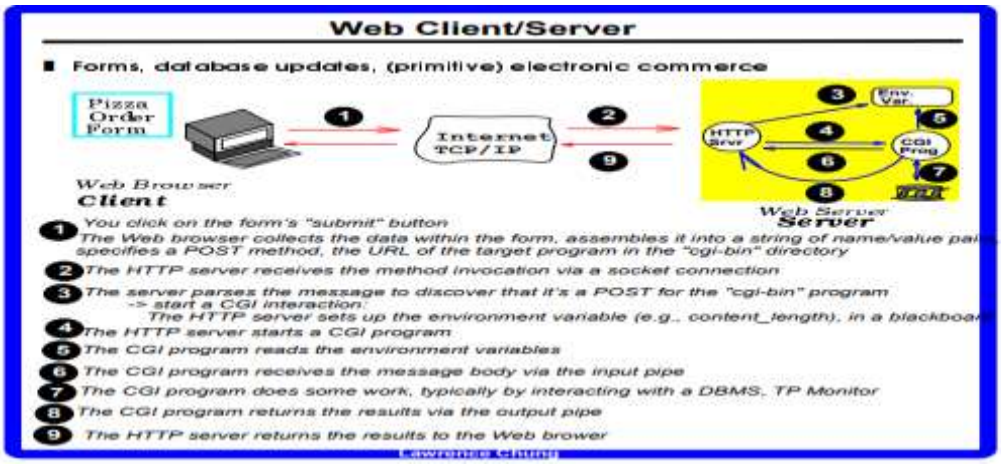
- **IP** - Internet Protocol

IP takes care of the communication between computers. It is responsible for addressing, sending and receiving the data packets over the Internet.

- **FTP** - File Transfer Protocol

FTP takes care of transmission of files between computers.

- A socket is line of plumbing established either client or server software that is used to move packets between client and server.
- A Packets is a fragments of data includes information about its origination and destination.
- Web Application communicates over Sockets using higher level protocol the HTTP (HyperText Transfer Protocol).
- HTTP is a protocol which allows the fetching of resources, such as HTML documents.
- It is the foundation of any data exchange on the Web and it is a client-server protocol, which means requests are initiated by the recipient, usually the Web browser.



SERVER SIDE PROCESSING WITH CGI

- CGI works by processing the data sent by the web browser to the web server.
- The data contained in the form fields are encoded by the browser using X-URL-encoding system before transmission.
- The data is sent by one of 2 methods:
  - Encoding the data an HTTP POST (Post request)
  - Passing the data to script file in URL(Get request).
- The Web Server receives this data and passes it to CGI application of processing.

1.2 COMPONENTS OF ASP.NET AND THE .NET FRAMEWORK

- The following are the components of asp.net and the .net framework:

- **Internet Information Server** → server software that deliver web content to the user browser.

- **Asp.net page** → HTML page that host static content and execute code. It act as the glue between the presentation layer and application services.
- **Common Language Runtime(CLR) and the ,Net framework Library** → CLR provide runtime environment for manage code and library provide components for developing applications.
- **Managed Components** → allow sharing of modules among many different applications.
- **Web Services** → provide components based functionality that access over network by any machine using variety of protocols.

1.3 Overview of Internet Information Server(IIS)

- IIS is Microsoft software for serving content on web.
    - IIS use **HTTP** → deliver web content
    - **FTP and SMTP** → send e-mail
- The functionality of IIS is extendable through the ISAPI(Internet Server Application Programmers Interface).
- ISAPI program come in two flavors:
    1. ISAPI Extension
    2. ISPAI Filters

1. ISAPI Extension

- Resemblance to traditional CGI program.